



Sepsis and Septic Shock

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg
- Pediatric hypotension is defined as SBP < 70 + (age in years x 2) mmHg

Signs & Symptoms:

- Trigger for sepsis guideline:
 - Known or suspected infection -and-
 - EtCO₂ < 30 or > 45
- AND 2 or more of the following:
 - Temp < 96.8 F or > 100.4 F
 - Heart rate > 95
 - SBP < 100
 - Respiratory rate > 20
 - Altered Mental Status

Differential:

- Arrhythmia
- Pulmonary embolism
- Anaphylaxis
- Drug intoxication
- Heat stroke
- Hypothermia
- Hypoglycemia
- Dehydration
- Stroke

Clinical Management Options

P	P	P	P	P	P	• Oxygen • BGL assessment • Keep patient warm • Treat wheezing, hypoxia, dyspnea, and pain as appropriate per COGs
L	L	L	L	L	L	
1	2	3	4	5	6	• 4 and 12 lead ECG • EtCO₂ • Adult - Acetaminophen
						• Pediatric - Acetaminophen
						• Vascular access • Isotonic Crystalloid fluid challenge: ◦ Adult: 30 ml/kg ◦ Pediatric: 20 ml/kg ◦ Newborn: 10 ml/kg
						• ECG interpretation • Norepinephrine infusion, titrate to MAP > 65 if not responsive to IV fluid • See Hypotension COG

Consult Online Medical Control As Needed

Pearls:

- Refer to drug formulary charts for all medication dosing for both adults and pediatric patients.
- Early septic patients often become hypothermic instead of developing fevers.
- Only about 3% of pediatric septic patients receive IV fluid resuscitation in the field, paramedics should treat pediatric sepsis per COG.
- Hypoglycemia is not uncommon in patients with sepsis, particularly those on beta blockers.
- Sinus tachycardia may be misinterpreted as SVT or A-fib. Sinus tachycardia > 150 bpm in the adult patient or > 180 in the pediatric patient may be seen with sepsis.